

Hallucination Detection in Graphs

The Issues-FS Research Brief:
Moving from rhetorical confidence
to structural provenance.

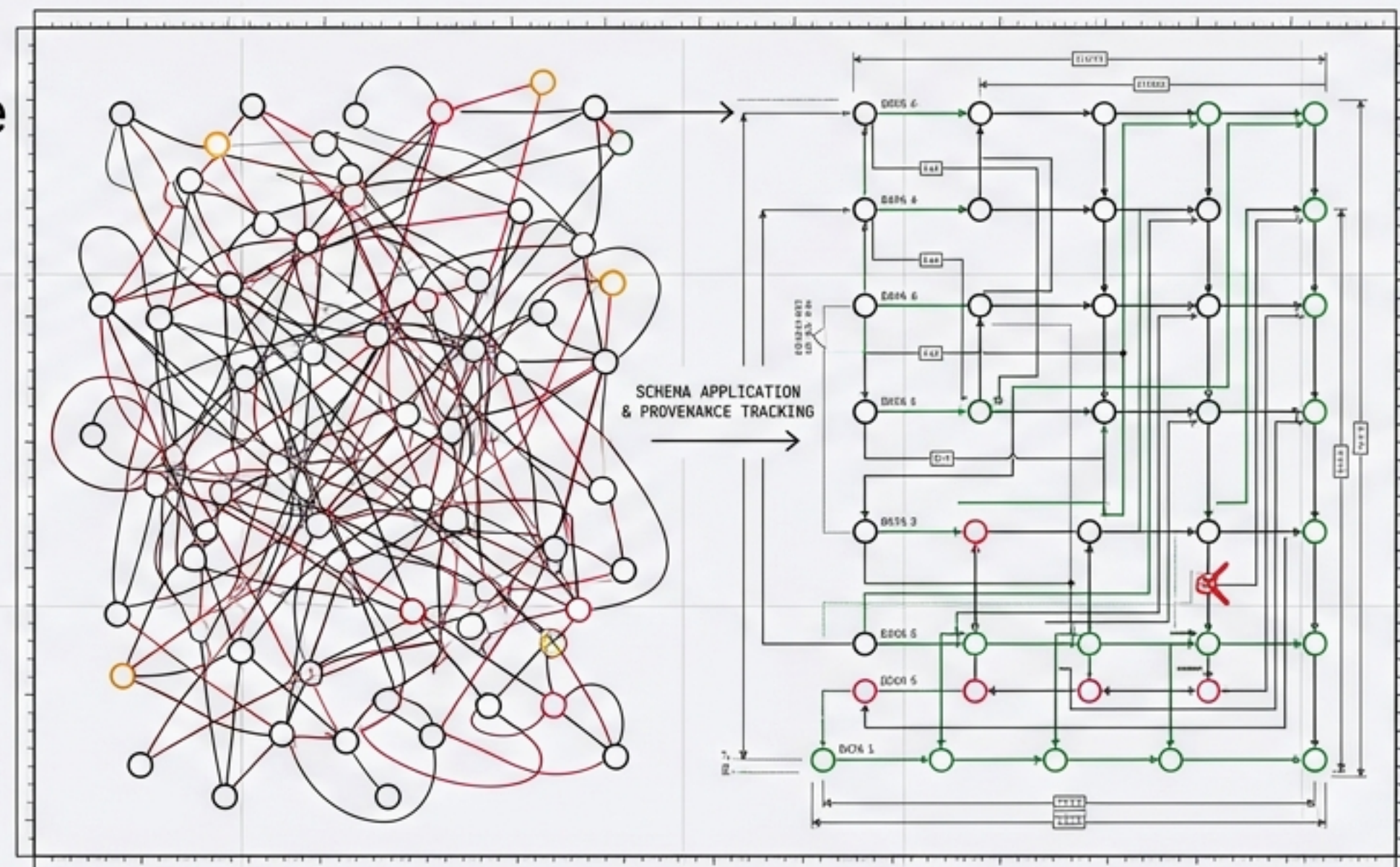
VERSION: v0.4.11

DATE: 19 Feb 2026

FROM: Human (Project Lead)

TO: Issues-FS Team

TYPE: Concept Exploration



The Structural Definition of Truth

We can flag, tag, query, and visualise hallucinations because Issues-FS is naturally suited for provenance traversal.



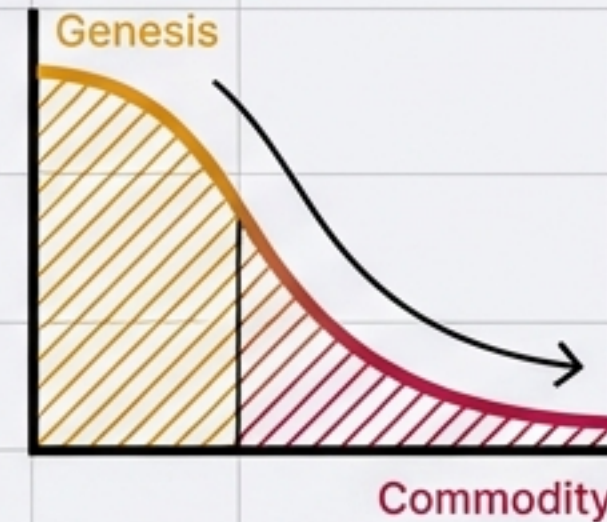
The Definition

A hallucination is a node with no valid provenance path to a trusted source. It is a 'dead end' in the graph.



The Spectrum

Truth is not binary. Information exists on a gradient from 'Fabricated' to 'Verified Fact' based on evidence depth.



The Tolerance

Hallucination is a feature in the 'Genesis' phase (creativity) but a critical bug in the 'Commodity' phase (reliability).



The Mechanics

We measure grounding via provenance depth, source count, credibility scores, and contradiction detection.

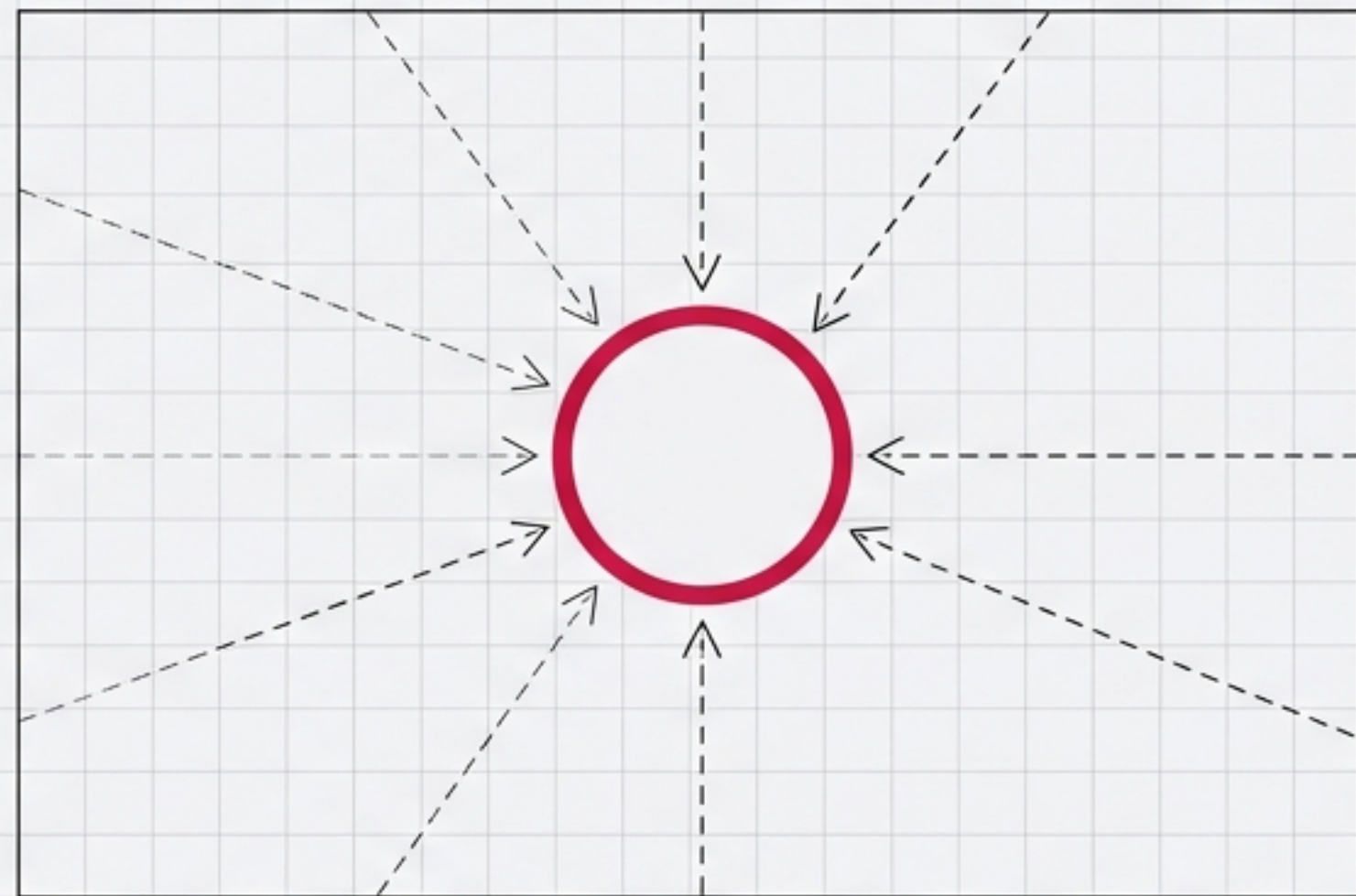
What is a Hallucination? (In Graph Terms)

THE LLM CONNOTATION

The LLM's response is delivered with high confidence, stating the information as fact, even if incorrect. It mimics the structure of truth.

Rhetorical Confidence: Sounding correct without substance.

THE STRUCTURAL REALITY



Provenance Dead End: A node with no incoming edges from a trusted source.

Definition: A piece of information for which we cannot find evidence to support it. This applies universally to LLMs, human opinions, and outdated data.

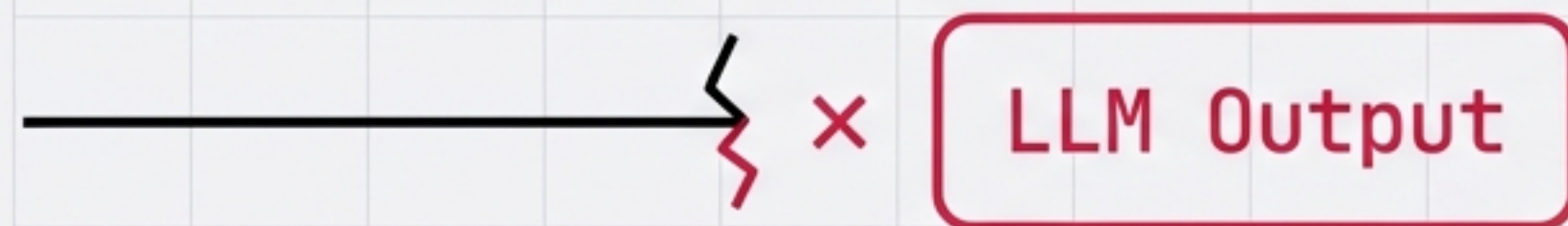
The Provenance Dead End

Detection is a matter of traversal.

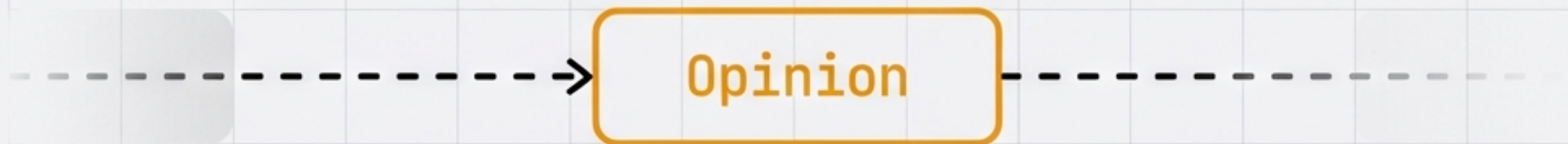
1. GROUNDED CHAIN (Verified)



2. UNGROUNDED CHAIN (Hallucination)



3. PARTIAL CHAIN (Ambiguous)



Issues-FS captures this naturally due to its fractal nature. We can zoom in on any claim to see the full tree.

The Spectrum Model



SCORING SIGNALS

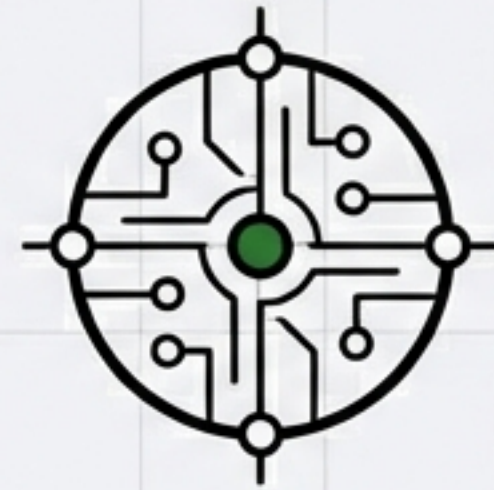
SIGNAL	METRIC
Provenance Depth	Deeper = More Grounded
Source Count	More Independent Sources = More Grounded
Contradiction Count	Zero Contradictions = More Grounded
Consistency	Graph Fit = More Grounded
Age	Fresher Timestamp = More Grounded

Source Credibility is Emergent

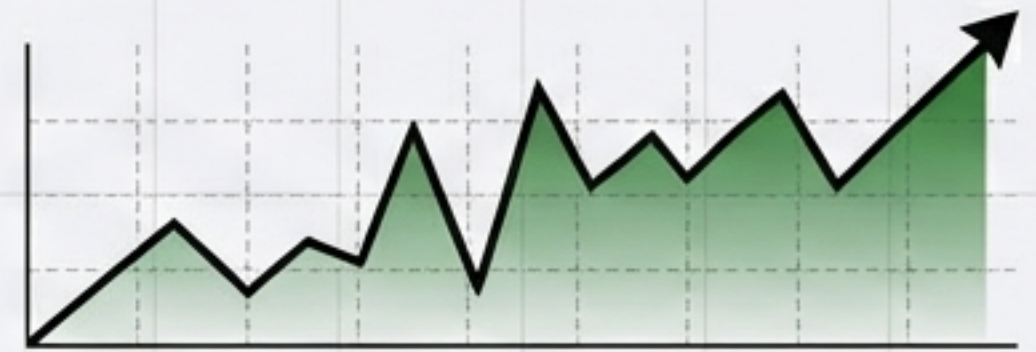
Credibility is computed from structural properties, not manually assigned.

CREDIBILITY CRITERIA

- [] **Consistency:** Does not flip-flop without reason.
- [] **Citations:** Provides provenance for its own claims.
- [] **Self-Correction:** Changes opinion only when evidence changes.
- [] **Verification:** Produces claims that are independently checkable.



SOURCE NODE ID: 08-AX



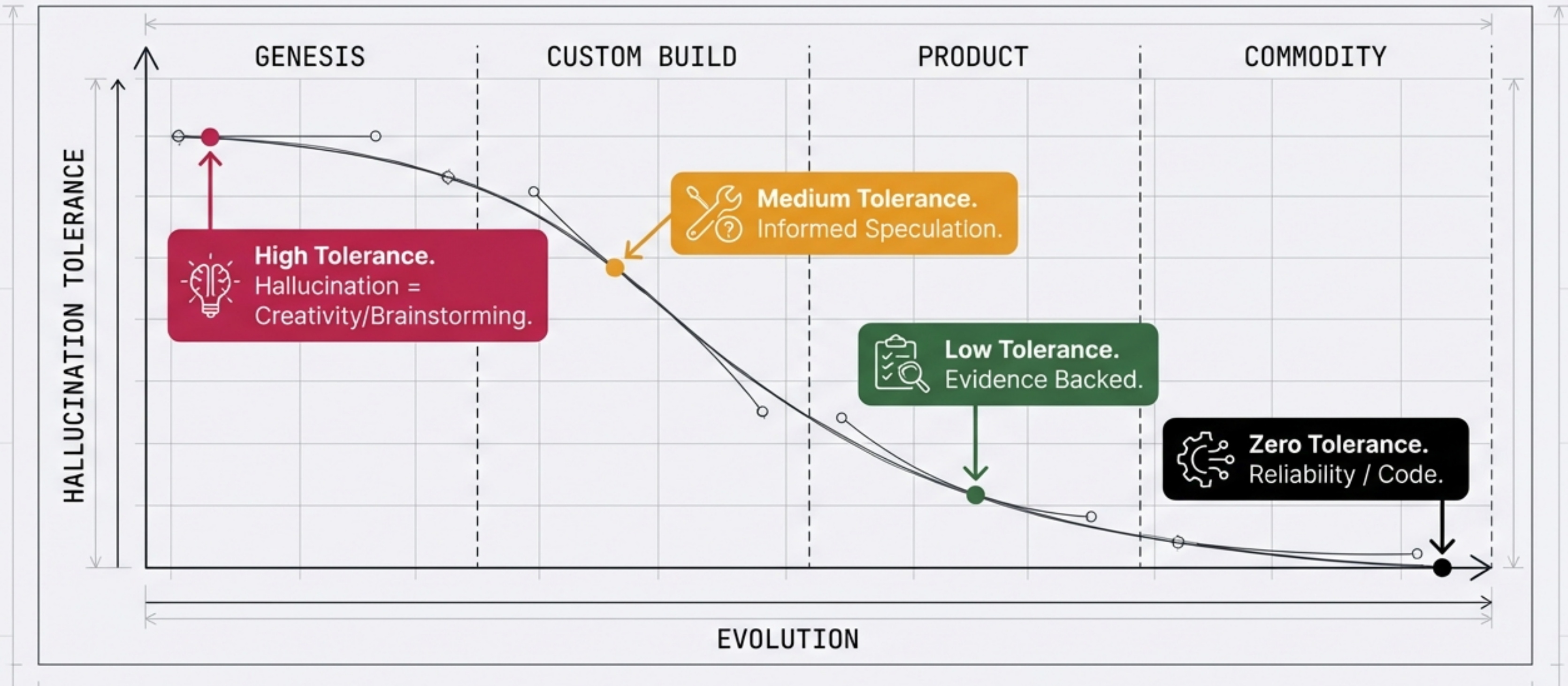
Consistency: **HIGH**

Citation Rate: **98%**

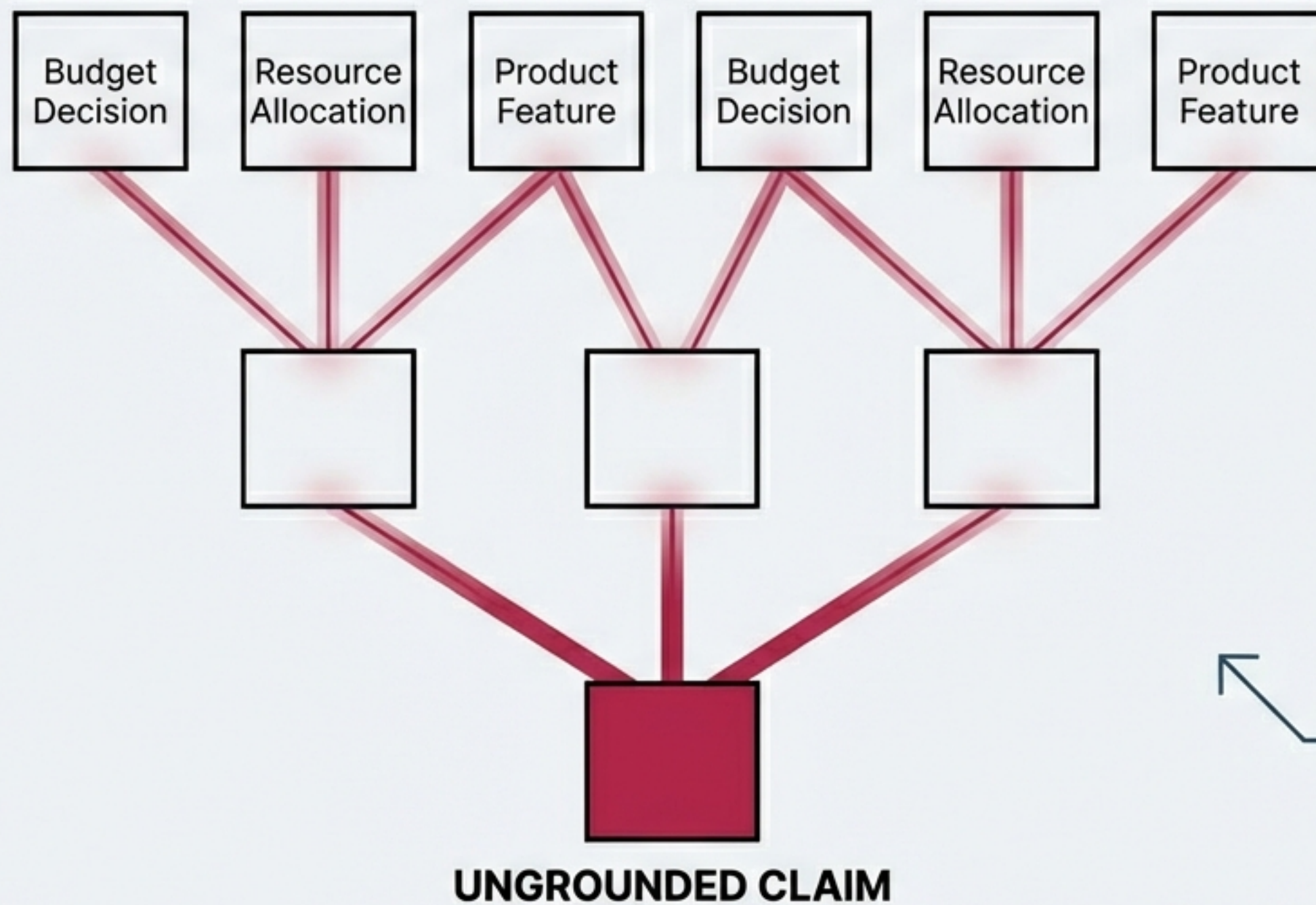
Correction Log: **ACTIVE**

→ **Insight:** A source that changes its position because of new evidence is **MORE** credible, not less.

Context Matters: The Wardley Evolution Scale



The 'Blast Radius' Query

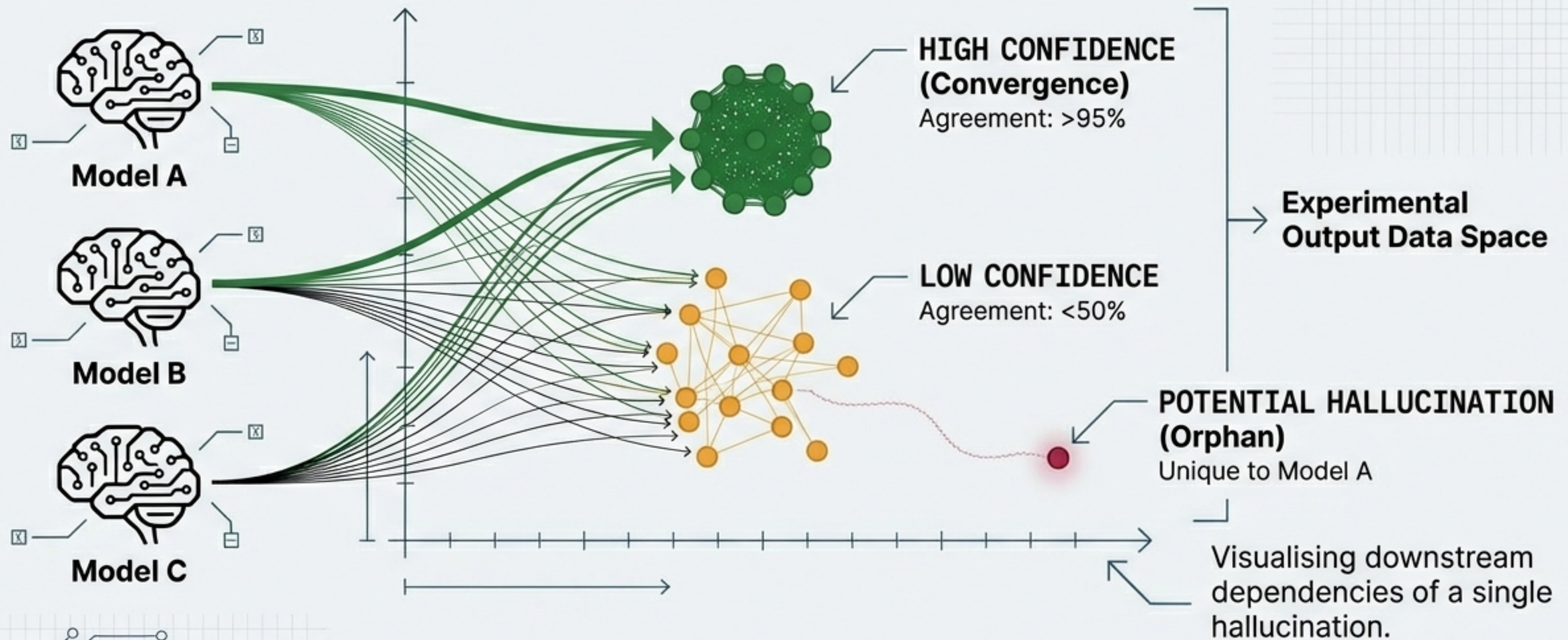


```
QUERY >  
MATCH (n:Claim {grounding:  
'low'})<-[:DEPENDS_ON*]-  
(impact)  
RETURN impact
```

Visualising downstream dependencies of a single hallucination.

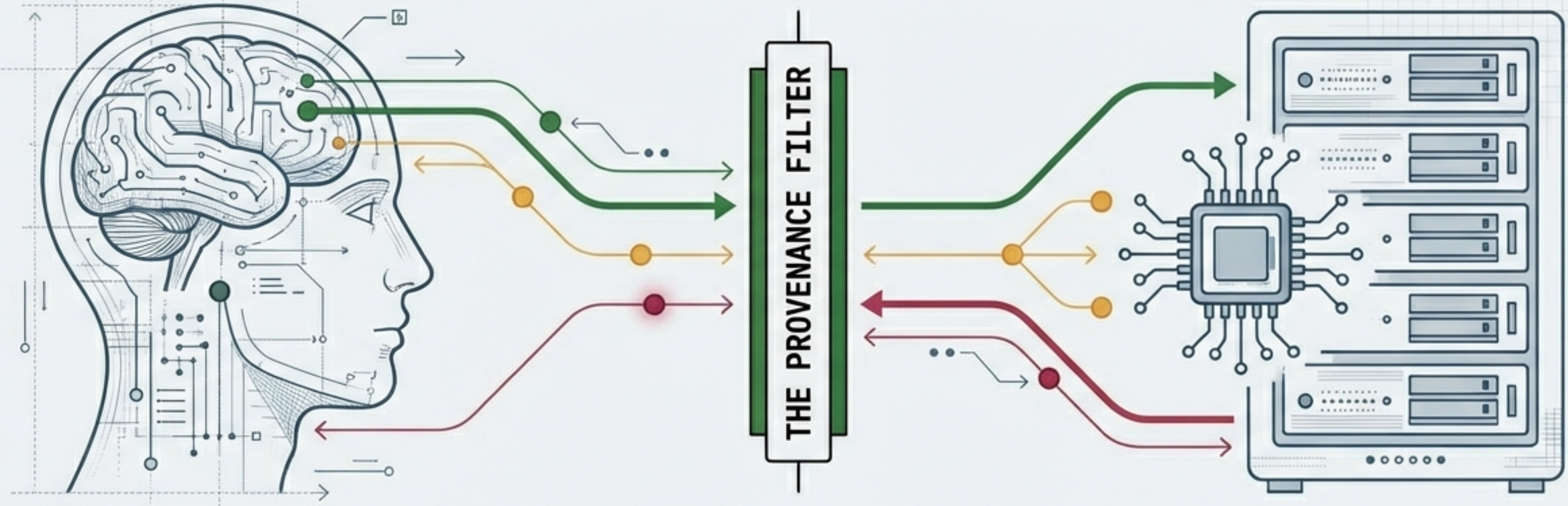
LLM Comparison as Experimental Data

Cross-model agreement as a proxy for truth.



People and Companies Hallucinate Too

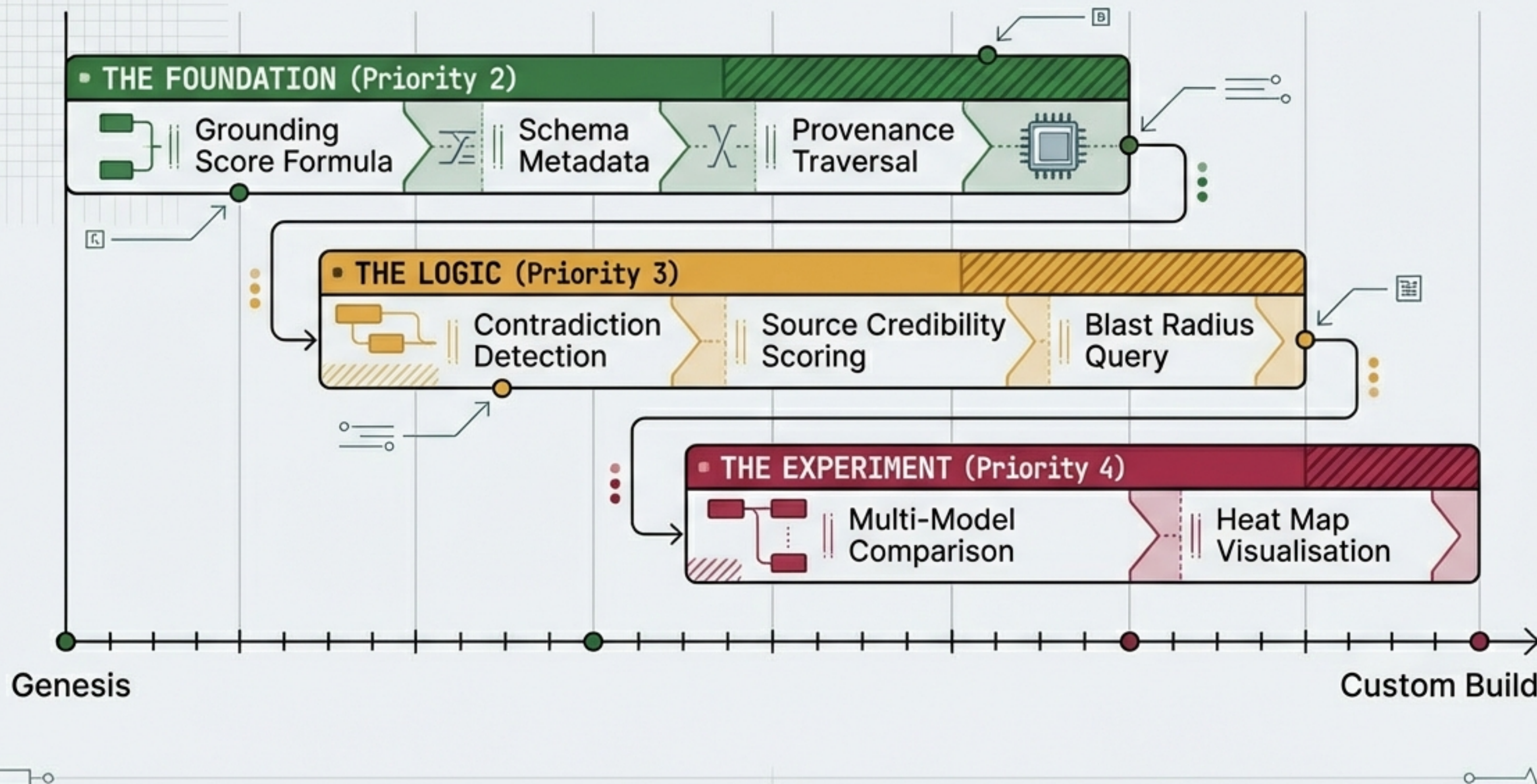
The graph treats all sources equally.



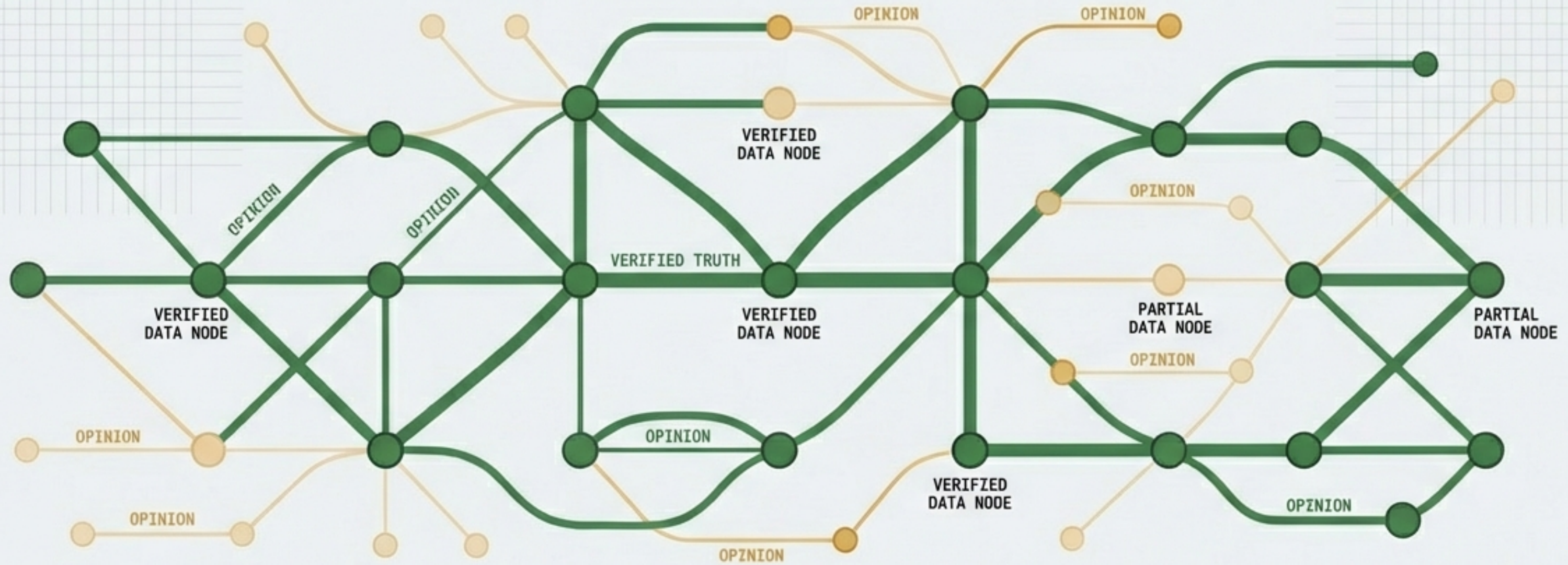
A human opinion without evidence = An LLM hallucination.

Examples: Unfounded earnings projections, Marketing hyperbole, Misremembered facts.

Implementation Roadmap



The Vision: Connected Intelligence



When we add connectivity to 'just words,' we get meaning. We enable the organisation to distinguish:

1. What we know (Grounded)
2. What we think (Partial)
3. What we made up (Fabricated)